

Internet Appendix for “*The Onset of LLMs: Leveling the Playing Field or Amplifying Elitism?*”

Appendix A1 Summary of literature review

Table A1 Literature on human-AI collaboration

Study	Tasks that AI assists with	Main Predictor	Consequence of Human-AI collaboration	Heterogeneous effect between workers
Abdel-Karim et al. (2023)	X-ray classification	Classification error rate	Decrease from 80.2% to 58.2%	-
Wang et al. (2024)	Medical code estimation	Time spent per chart	Time is reduced by 1.17minutes (11.29% in the pretreatment period) per medical chart	Significant improvements among more task experienced workers but more junior workers
Ge et al. (2021)	Loan risk estimation	Loan return rate	One unit increase of RA usage adjustment results in 0.2% decrease of average completed loan return or 0.1% decrease of average monthly return	-
Bauer et al. (2023)	Investment behavior estimation Real estate price estimation	Investment accuracy and recall Relationship between posterior and prior beliefs about attribute contribution to apartment price	The accuracy (recall) decreases in XAI provided group by 8.9% (11.0%) XAI leads to mental adjustments that are subject to conformation bias	-

Table A2 Literature on human-LLM collaboration

Study	Tasks that LLMs assist with	Main Predictor	Consequence of Human-LLM collaboration	Heterogeneous effect between workers
Brynjolfsson et al. (2025)	Conversational customer-support services	Resolution per hour (RPH)	LLM deployment on average increases RPH by 0.30 chats	Workers with less than one month of tenure improve by 0.7 RPH, while no effect for agents with more than a year of tenure Workers in the lowest skill quintile see an increase of 0.5 in RPH, while no significant change in most skilled workers
Chen and Chan (2024)	Creative work such as designing	Ad clicks	Nonexperts benefit from LLMs only in sounding board mode; experts experience performance declines regardless of LLM usage mode	Nonexperts
Noy and Zhang (2023)	Professional writing tasks such as press releases	Time taken and evaluator grades	Time taken in the treatment group on the posttreatment task dropped by 40%. Average evaluator grades in the treatment group increased by 18%	Inequality between workers decreased, evidenced by a lower task performance correlation in the treatment group than in the control group

In summary, Prior research has predominantly focused on traditional AI systems, while emerging studies on Human–LLM interaction largely concentrate on basic tasks such as writing, often within controlled experimental settings. However, the role of LLMs in supporting human decision-making in complex, domain-specific tasks remains underexplored. Given that LLMs represent a substantial advancement in AI capabilities, it is critical to investigate their influence in expert-driven contexts. This study addresses these gaps by empirically evaluating the impact of LLM adoption on financial analysts’ performance using large-scale real-world data.

Appendix A2 Sample selection and data

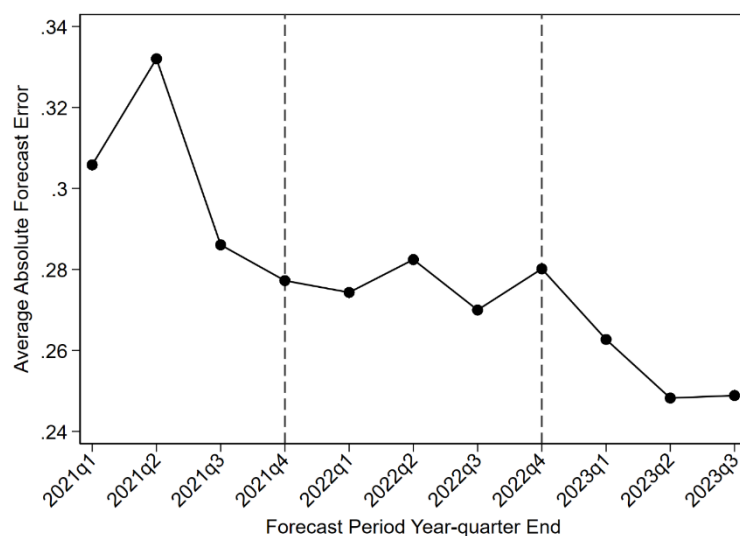
Table A3 Sample selection and distribution

Panel A: Sample selection process				
Sample selection criteria	Analyst forecast observations			
One-quarter ahead forecasts of Forecast quarter ends over 2021Q1 to 2023Q3	391,478			
Drop: Forecasts are not issued after the prior quarter’s earnings announcement and before the current quarter’s announcement	(1,682)			
Drop: Forecasts issued per analyst for the firm-quarter are not the most recent	(104,972)			
Drop: Forecasts issued by analysts with less than five observations over the full sample period.	(443)			
Drop: Analysts who do not issue at least one forecast for the firm during both the pre-GPT period and the post-GPT period	(98,385)			
Drop: Forecasts with missing data to calculate analyst characteristics	(380)			
Drop: Firm-quarter pairs for which less than two analysts issued a forecast	(13,708)			
Final sample	171,908			
Panel B: Sample distribution				
Fiscal-quarter end	No. Analysts	No. Forecasts	No. Brokers	No. Firms
2021Q1	1,380	12,146	148	1,810
2021Q2	1,447	12,690	155	1,887
2021Q3	1,482	13,464	155	1,965
2021Q4	1,552	14,498	165	2,102
2022Q1	1,614	15,616	161	2,244
2022Q2	1,674	16,549	161	2,355
2022Q3	1,733	18,023	164	2,490
2022Q4	1,732	18,257	162	2,528
2023Q1	1,733	18,084	158	2,499
2023Q2	1,682	16,675	159	2,350
2023Q3	1,599	15,906	155	2,269
Total	17,628	171,908	1,743	24,499

Note: This table presents the sample selection and distribution. Panel A summarizes the sample selection process. Panel B reports the distribution of the sample by quarter.

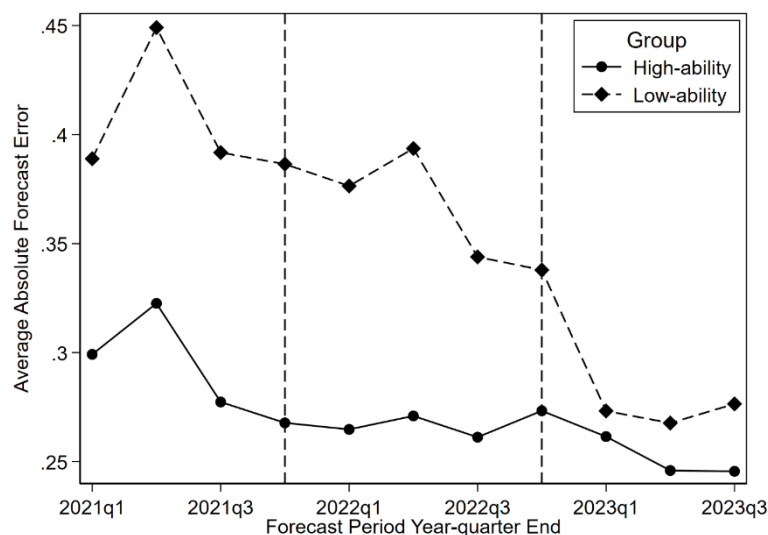
Appendix A3 Descriptive results

Figure A1 Time trend of absolute forecast error



Note: This figure shows the time distribution of absolute forecast error (*AFE*). The vertical axis represents the average absolute forecast error for each quarter. The horizontal axis denotes the forecast period ending by year-quarter, ranging from 2021Q1 to 2023Q3.

Figure A2 Time trend of absolute forecast error by group



Note: This figure shows the time distribution of absolute forecast error (*AFE*) for the treatment and control groups, respectively. The dashed line with square points represents the control group (*Treat*=0), while the solid line with circle points represents the treatment group (*Treat*=1). *Treat* is a dummy variable that equals one for forecasts issued by analysts whose performance scores are always below the firm-quarter median throughout the pre-GPT period, and zero otherwise. The vertical axis represents the average absolute forecast error for each quarter by group. The horizontal axis denotes the forecast period ending by year-quarter, ranging from 2021Q1 to 2023Q3.

Appendix A4 Parallel trend test

Table A4 Parallel trend test

	PMAFE
D'Treat	0.484*** (17.15)
D'Treat*2021Q2	-0.039 (-1.20)
D'Treat*2021Q3	-0.013 (-0.40)
D'Treat*2021Q4	0.015 (0.41)
D'Treat*2022Q1	0.043 (1.38)
D'Treat*2022Q2	0.077** (2.25)
D'Treat*2022Q3	0.029 (0.89)
D'Treat*2022Q4	-0.117*** (-3.62)
D'Treat*2023Q1	-0.305*** (-9.65)
D'Treat*2023Q2	-0.294*** (-8.90)
D'Treat*2023Q3	-0.307*** (-9.64)
Control	YES
Cluster	Firm-analyst
N	171,908
Adj.R2	0.036

Note: This table presents the results of the parallel-trend test. The dependent variable is *PMAFE*. Prefix “D” denotes variables are demeaned within firm-quarter. *Post* is omitted after being demeaned within firm-quarters. All variables are defined in the Appendix. T-statistics are given in parentheses based on standard errors double clustered at the firm-analyst level. *, **, and *** denote significance based on two-tailed t-tests at the 10%, 5% and 1% levels, respectively.

Appendix A5 Robustness tests

We conduct a series of tests to assess the robustness of our findings across various research settings. The dependent variable in model (1) is the adjusted $D'AFE$ (i.e., $AFE-MAFE$), normalized by $MAFE$. This transformation reduces heteroscedasticity and allows the variable to be interpreted as the analyst's fractional forecast error relative to the average of all analysts' absolute forecast errors covering the same firm in the same quarter (Clement, 1999). In our initial robustness test, we directly employ $D'AFE$ as an alternative measure (Hilary & Menzly, 2006). Column (1) of Table A2 presents the results of using $D'AFE$. We find that the estimated coefficient on $D'Treat*Post$ in column (1) is negative and significant. These findings indicate that our results are robust, showing consistent patterns even when a different measure of the dependent variable is employed.

In our second robustness test, we explore the sensitivity of our results to the selection of forecasting horizons. Specifically, we impose a criterion that forecasts must be made by analysts at least one month prior to the current quarter's earnings announcement. This adjustment yields a sample of 156,823 forecasts. We replicate the regression specified in model (1), with the results detailed in column (2) of Table A2. Although the magnitude of the coefficient on $D'Treat*Post$ is smaller compared to the baseline results, its sign remains significantly negative. Consequently, our findings are robust, affirming their validity under varying forecast horizon selections.

In our research design, the number of forecasts issued by the treatment group (low-quality analysts) is substantially smaller than those issued by the control group (high-quality analysts). To address this imbalance, we conduct a propensity score matching method as our third robustness test. Specifically, for cross-sectional observations of each fiscal quarter, we calculate the propensity scores for treatment and

control groups by using analyst characteristics defined in model (1). Each treatment observation (forecasts issued by low-quality analysts) is then matched to a control observation (forecasts issued by high-quality analysts) based on the nearest propensity score. This matching process ensures that the number of forecasts issued by both high-quality and low-quality analysts is balanced in each quarter, and it also mitigates the potential impact of differences in analyst characteristics between the two groups. After matching, we obtain a sample of 31,993 observations and re-estimate model (1) using this refined sample. The results, presented in column (3) of Table A2, show that the coefficient on $D'Treat*Post$ remains negative and significant, indicating that our main findings are robust even after addressing the imbalance through propensity score matching.

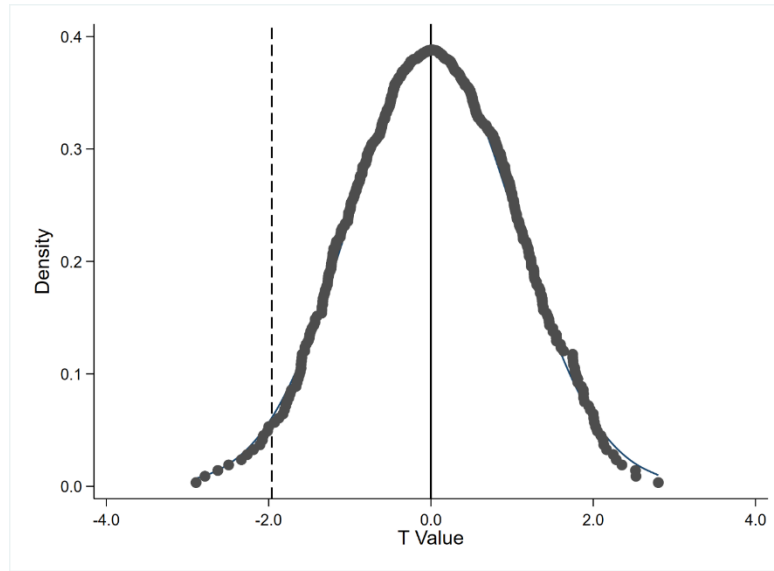
In our model (1), we adopt the methodology of Clement (1999) by demeaning all variables by their respective firm-quarter means to control for firm-quarter fixed effects. Similarly, prior studies have estimated models using raw variables while incorporating firm-quarter paired fixed effects to achieve comparable objectives (Michaely et al., 2024). In our final robustness tests, we align with the empirical specifications of these studies and replicate our model (1) using unadjusted variables. We examine two dependent variables: the absolute forecast error (AFE) and the absolute forecast error scaled by the stock price two days prior to the current quarter's earnings announcement date (AFE/P). To control for time-invariant brokerage characteristics that may influence analyst forecast accuracy, we also include brokerage fixed effects in our analysis. The results, presented in columns (4) and (5) of Table A2, reveal that the coefficients on $Treat*Post$ in these two columns are both significantly negative. The coefficient in column (4) has a comparable magnitude to that observed in columns (1). These findings corroborate the robustness of our main results across different empirical methodologies.

Table A5 Robustness tests

	(1)	(2)	(3)	(4)	(5)
	D'AFE	PMAFE	PMAFE	AFE	AFE/P
Treat	0.111*** (10.89)	0.460*** (28.75)	0.389*** (30.76)	0.110*** (10.93)	0.004*** (6.34)
Treat*Post	-0.075*** (-9.62)	-0.282*** (-18.33)	-0.246*** (-12.71)	-0.074*** (-9.60)	-0.003*** (-3.44)
Top10	-0.003 (-0.99)	0.002 (0.38)	0.015 (1.32)	-0.014** (-2.16)	-0.001 (-1.05)
Horizon	0.000*** (8.07)	-0.001*** (-6.92)	0.001*** (3.66)	0.000*** (8.16)	0.000** (2.22)
Companies	0.000 (1.05)	0.000 (1.03)	-0.001 (-1.47)	0.000 (1.62)	0.000 (1.30)
Industries	0.001 (1.47)	0.004** (2.37)	0.008** (2.19)	0.001 (1.13)	0.000 (0.32)
Fexperience	-0.000 (-0.53)	0.000 (0.52)	-0.001** (-2.25)	-0.000 (-0.63)	0.000* (1.94)
Gexperience	-0.000 (-0.86)	0.000 (0.05)	0.000 (0.27)	-0.000 (-0.99)	-0.000*** (-2.75)
Model	Mean-adjusted	Mean-adjusted	Mean-adjusted	Firm-quarter FE	Firm-quarter FE
Cluster	Firm-analyst	Firm-analyst	Firm-analyst	Firm-analyst	Firm-analyst
N	171,908	156,823	31,993	171,908	171,908
Adj.R2	0.011	0.033	0.081	0.853	0.940

Note: This table presents the results of robustness checks. *Post* is omitted after being demeaned within firm-quarters or controlling for firm-quarter fixed effects. Column (1) uses the firm-quarter-demeaned *AFE* as an alternative dependent variable. Column (2) presents the results of sample restricted to earnings forecasts with a horizon of at least 30 days. Column (3) shows the results after PSM matching. Column (4) and column (5) are estimated through raw variables and firm-quarter paired fixed effects. The dependent variable in column (4) is unadjusted *AFE*, while that in column (5) is unadjusted *AFE* divided by the stock price two trading days prior to the current quarter's earnings announcement date. T-statistics are given in parentheses based on standard errors double clustered at the firm-analyst level. *, **, and *** denote significance based on two-tailed t-tests at the 10%, 5% and 1% levels, respectively.

Figure A3 Placebo test of pseudo-treatment analysts



Note: This figure presents the distribution of the T-statistics of the estimated coefficients on *Treat*Post* from five hundred Pseudo-treatment regressions. T-statistics are based on standard errors that are double-clustered at the firm-analyst level. The horizontal axis denotes the corresponding T-values, while the vertical axis represents the density of the T-values.

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